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Effects of process parameters of rotary tablet press on die filling behavior during mini-tablet production: Comparison with conventional tablet

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ABSTRACT

This study focuses on the effects of process parameters of rotary tablet press on die filling behavior during the mini-tablet production. Fill cam type, adjusted fill depth, and disc rotational speed were modulated, and mean tablet weight, mean powder fill level, and relative standard deviations for tablet weights were evaluated. To examine the differences between the production processes of mini-tablets and conventional tablets, typical-sized tablets were studied as a reference. When producing conventional tablets, excessively deep fill cam and high disc rotational speed negatively affected die filling efficiency and weight variation. In contrast, for mini-tablets, deep fill cam and high disc rotational speed were suitable for efficient and reproducible die fill. These were closely related to the die retention time under powder bed and the degree of powder suction. Since die filling for mini-tablet production was significantly different from conventional tablet, understanding its characteristics is necessary to ensure tablet quality.

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1. Introduction

Generally, mini-tablets are defined as tablets with a diameter of less than 3 mm [1]. Owing to their small size, they have advantages over conventional tablets. Instead of swallowing one big tablet, patients can take several units of mini-tablets [2]. Many clinical studies [3–5] showed the potentials of mini-tablets for improving medication adherence in pediatric patients who experience difficulty in swallowing tablets of typical size. Even adults prefer taking mini-tablets over conventional tablets [6]. In addition, mini-tablets are suitable for individual dosing which has recently been the focus of much research. The mini-tablet system enables more accurate dosing compared with tablet splitting [7] and provides flexibility for dose adjustments according to the symptoms [8]. Moreover, mini-tablets can be used to prepare single dosage forms for various purposes, such as combined drug release [9–12], physical separation of incompatible ingredients [13], and delivery of multiple compounds at a time [14]. Although there are other multi-particulate systems including pellets and granules, mini-tablets have the advantages. They show more uniform size, shape, mass, porosity, and smooth surface [1], while needing less coating thickness for sustained-release [15,16].

The distinct difference between the production process of conventional tablets and mini-tablets is that the size of the die opening for mini-tablets is much smaller than that for conventional tablets. Therefore, a detailed study on die filling behavior during mini-tablet production is necessary. However, most research on mini-tablets mainly focused on the feasibility of producing small-sized tablets [17,18] or their applications including oral disintegration [19,20], solubilization of poorly soluble drugs [21,22], and sustained release [23,24]. It was not until recently that researchers began focusing on the fundamentals of the manufacturing process of mini-tablets. In their studies, a custom-made instrument [25,26] and compaction simulator [27] were used to understand die filling behavior at small die opening. However, the actual powder feeding process occurring on a rotary tablet press cannot be easily simulated because it is affected by various factors, concurrently. Therefore, studies using practical equipment are still needed. Although a few studies utilizing rotary tablet press were conducted to verify the methods for evaluating variations in die filling behavior [28] or elucidate the effects of feed frame parameters and turret speed [29], the former did not cover much on the effects of adjusting the process parameters, while the range of turret speed used in the latter had no significant effects on the weight of mini-tablets.

In the circumstances above, detailed investigation on the various process parameters of rotary tablet press regarding the mini-tablet production was required. From this aspect, this study aimed to examine the effects of cam type, adjusted fill depth, and rotational speed on die filling

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behavior during mini-tablet production on a rotary tablet press. The production processes of the conventional-sized tablets were also studied as reference for systematic comparison.

2. Materials and methods

2.1. Materials

Microcrystalline cellulose (Comprecel® M101, Mingtai Chemical Co., Ltd., Taiwan), lactose monohydrate (Granulac® 200, Meggle, Germany), hydroxypropyl cellulose (Klucel® LF, Ashland, USA), and magnesium stearate (Duksan, Republic of Korea) were used for this study. Purified water was obtained from an ultra-pure water system (MR-RU890, Mirae St Co., Ltd., Republic of Korea).

2.2. Preparation of granules

As shown in Table 1, microcrystalline cellulose, lactose monohydrate, and hydroxypropyl cellulose were accurately weighed to achieve a total amount of 3 kg and blended using an IBC blender (PcM-IBC10, GEA Pharma Systems, Belgium) (10 rpm, 10 min, total 100 revolutions). Next, wet granulation was performed with a high speed kneading granulator (Triple Master TMGV-05, Shinagawa Machinery Works, Japan), which was found to be suitable for manufacturing micro sized granules in our previous study [30]. The batch size was 150 g and purified water was used as a binding liquid. The standard type of impeller and chopper were equipped, and granulation was conducted by the two-step method for a total of 5 min. The process conditions were selected based on the results of preliminary study (Supplementary material 1). For the first 3 min of the process, the conditions were as follows: revolution rpm, +50; impeller rpm, −500; chopper rpm, +500. For the last 2 min, the conditions were: revolution rpm, +50; impeller rpm, −1000; chopper rpm, +1000 (+, clockwise; −, anticlockwise). During the first step of the granulation process, 60 g of purified water was sprayed to the powder bed through the spray nozzle (diameter, 1 mm; angle, 45°) with an air pressure of 0.025 MPa. By adjusting the flow rate of the peristaltic pump, it took approximately 2 min for all fluids to enter. After the granulation process, the wet mass was put on a tray and dried in a convection oven at 60 °C for more than 4 h to achieve granule moisture content of approximately 1%. Next, the granules were milled using a conical screen mill (U3, Quadro®, Canada) equipped with a square bar impeller (1607, Quadro®) and a graterholed screen with hole size of 1 mm (40G, Quadro®). The rotational speed of the impeller was set to 4500 rpm and samples were manually added into the conical screen mill to a volume of not less than 50% of the volume of the bowl to ensure consistent functioning. A plastic lid with a hopper was used to prevent samples escaping from the bowl. The process above was repeated eight times, and all products were mixed to obtain 1.2 kg of dried granules. At last, they were lubricated with magnesium stearate using a shaker mixer (Turbula® T2F, Willy A. Bachofen AG Maschinenfabrik, Switzerland) at 49 rpm for 5 min, so that the amount of lubricant accounts for 1% of the final blends. Magnesium stearate was sieved through a 355-μm aperture sieve before use. The final blends were stored at 25 °C and 15.9% R.H. for more than 24 h prior to following experiments.

Table 1
Compositions and proportions of the dry mixture for wet granulation.

	Compositions	Proportions (%)
Intra granular	MCC (Comprecel® M101)	48.5
	Lactose monohydrate (Granulac® 200)	48.5
	HPC (Klucel® LF)	3
Total	–	100

2.3. Characterization of granules

The prepared granules were characterized for suitability of mini-tablet production. Particle size distribution and flow properties were evaluated as follows. All tests were conducted in triplicate.

2.3.1. Particle size distribution of granules

Particle size distribution of the prepared granules was examined using the sieve analysis. All granules were passed through a 710-μm aperture sieve before the tests. A total of 10 g granules was accurately weighed and put on the top of a series of sieves (lid, 425-, 355-, 300-, 250-, 200-, 150-, 106-, 75-μm aperture sieves, and receiver). Next, the sieves were gently shaken for 1 min (manually) to divide the granules according to their size fractions. Volumetric diameters were deduced by converting fractional size distribution to cumulative size distribution.

2.3.2. Flow properties of granules

Flow properties of the prepared granules were evaluated using the compendial methods. Bulk density of granules was measured by gently pouring approximately 36 g of granules into the 100-mL plastic volumetric cylinder. Next, the cylinder was manually tapped at a rate of 250 taps/min and height of 1 cm. The powder density after the 1250 taps was regarded as the tapped density. The compressibility index and Hausner ratio were derived from the results of bulk and tapped density. The following equations were used for the calculations.

$$\text{Bulk density (g/mL)} = \frac{\text{Mass of granules (g)}}{V_O\ (\text{mL})}$$

$$\text{Tapped density (g/mL)} = \frac{\text{Mass of granules (g)}}{V_F\ (\text{mL})}$$

$$\text{Compressibility index} = \frac{V_O\ (\text{mL}) - V_F\ (\text{mL})}{V_O\ (\text{mL})} \times 100$$

$$\text{Hausner ratio} = \frac{V_O\ (\text{mL})}{V_F\ (\text{mL})}$$

where V_O and V_F represent the unsettled apparent volume and tapped volume of granules, respectively.

The angle of repose was obtained by gently pouring 20 g of granules through the funnel (inner diameter of 8 mm) to a plastic cylinder with fixed diameter base (diameter, 55 mm; height, 12 mm). The distance between the tip of the funnel and plastic cylinder was fixed to 4 cm to minimize the impact of the falling powder on cone formation. The following equation was used to calculate angle of repose (α, °). The volume of poured powder was large enough to occupy the entire circumference of the bottom cylinder.

$$\tan(\alpha) = \frac{\text{Height of the cone (mm)}}{0.5 \times \text{Diameter of the plastic cylinder (mm)}}.$$

2.4. Die filling study

In this study, we investigated the die filling phenomenon during the manufacture of tablets with two different sizes. One tablet was small but regarded to be of typical size (5.5 mm, hereinafter referred to as conventional tablets), whereas the other was tiny of non-typical size (2 mm, hereinafter referred to as mini-tablets). Throughout the experiments, process parameters such as the type of fill cam (Fig. 1, type B and C for conventional tablets, and type A and B for mini-tablets), adjusted fill depth (6.55 and 7.96 mm for conventional tablets; 2.98 and 3.96 mm for mini-tablets), and disc rotational speed (15, 30, 45, 60, and 75 rpm for both tablets) were modulated to evaluate their effects on die

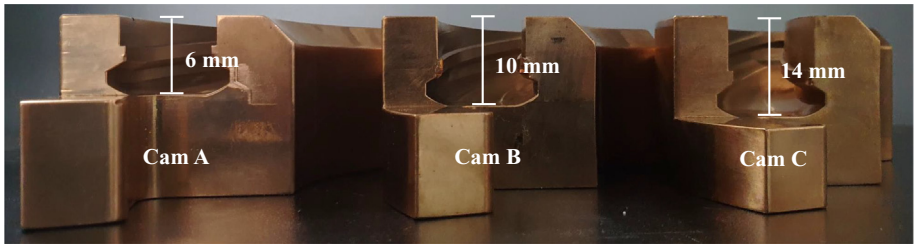


Fig. 1. Comparison of fill cams. Maximum fill depth: 6 mm for cam A, 10 mm for cam B, and 14 mm for cam C.

filling. A force feeding system equipped with three feeder paddles was used to provide powder into the die fill region and its rotation speed was set to 50% of the disc rotational speed to maintain constant number of paddle passes over the die cavity. To prevent severe damage to the equipment and tooling, the distance between the upper and lower punches was set to not less than 1 mm. Regarding tablet compression, no pre-compression stage was applied, and the depth of upper punch penetration was fixed to 3 mm, whereas the position of the lower punch was adjusted. All processes were conducted at 25 °C and 15.9% R.H.

2.4.1. Die filling study of conventional tablets (5.5 mm)

Conventional tablets were produced using a rotary tablet press (GRC-15S, Sejong Pharmatech, Republic of Korea) which contained a total of 15 stations. Three sets of 5.5 mm-diameter round punches with concave faces and associated dies (Euro D type, Young Chang Punch Co., Ltd., Republic of Korea), as shown in Fig. 2a, were mounted at stations number 1, 6, and 11 to achieve equal intervals between each station. As presented in Fig. 3a and Table 2, two types of fill cam (type B, fill depth of 10 mm; type C, fill depth of 14 mm) and two different adjusted fill depths (6.55 and 7.96 mm) were applied. When the adjusted fill depth was 6.55 mm, the final tablet thickness was set to 2.55 mm regardless of the cam type. In the same manner, with the adjusted fill depth of 7.96 mm, the final tablet thickness was set to 3.04 mm. Disc rotational speed was adjusted from 15 to 75 rpm, simultaneously.

2.4.2. Die filling study of mini-tablets (2 mm)

To produce mini-tablets, three sets of 2 mm diameter round 19-tip punches with concave faces and undercut dies (Euro D type, HD Pharmatool, Republic of Korea), as shown in Fig. 2b, were mounted on a rotary tablet press (GRC-15S, Sejong Pharmatech), as mentioned above. As presented in Fig. 3b and Table 2, two types of fill cam (type A, fill depth of 6 mm; type B, fill depth of 10 mm) and two different adjusted fill depths (2.98 and 3.96 mm) were applied. The final tablet thickness was set to 1.04 and 1.21 mm for adjusted fill depth of 2.98 and 3.96 mm, respectively. Furthermore, the effect of disc rotational speed on die filling was evaluated. Representative photographs of conventional tablet and mini-tablet are shown in Fig. 2c.

2.4.3. Weight measurement of tablets

After tablet production, 30 tablets were randomly selected and weighed individually to obtain their mass distribution. All samples were stored at 25 °C and 15.9% R.H. for more than 24 h before analysis.

2.4.4. Calculation of powder fill levels

The obtained weight of tablets was converted to powder fill level using the following equation. A powder fill level of 1 indicates that the actual fill mass of the powder inside the die cavity (namely the tablet weight) corresponds to the theoretical fill mass of the powder which is calculated from the granule bulk density and volume of the die cavity. The drawings provided from the punch manufacturer were used to calculate volume of die cavity (Supplementary material 2).

$$\text{Powder fill level} = \frac{\text{Actual fill mass of powder (g)}}{\text{Theoretical fill mass of powder (g)}}$$

$$\text{Theoretical fill mass of powder (g)} = \text{Granule bulk density (g/mL)} \times \text{Volume of the die cavity (mL)}$$

2.5. Statistical analyses

The significance of the results was analyzed with a software (SPSS® release 21.0, IBM, USA). For comparing two groups, independent two-sample *t*-test combined with the Levene's test for equality of variances was performed. For comparing three or more groups, one-way analysis of variance followed by Tukey's honest significant difference test was conducted.

3. Results and discussion

3.1. Granule properties

As reported by previous studies [27,31], particles sized larger than one third of die diameter might cause clogging of the die. From this aspect, the particle size distribution of the prepared granules was acceptable (Table 3). In addition, as evaluated by the criteria specified in the USP38-NF33, the flow properties were suitable for the tableting process in this study (Table 4). The particle size distribution of granule obtained from the feed frame after the tableting process did not differ



Fig. 2. The punches and dies used for producing (a) conventional tablets (5.5 mm) and (b) mini-tablets (2 mm). (C) Representative tablets produced (cam B, rotational speed of 30 rpm, adjusted fill depth of 7.96 mm for conventional tablet; cam B, rotational speed of 30 rpm, adjusted fill depth of 3.96 mm for mini-tablet) beside a Korean coin (diameter of 24 mm).

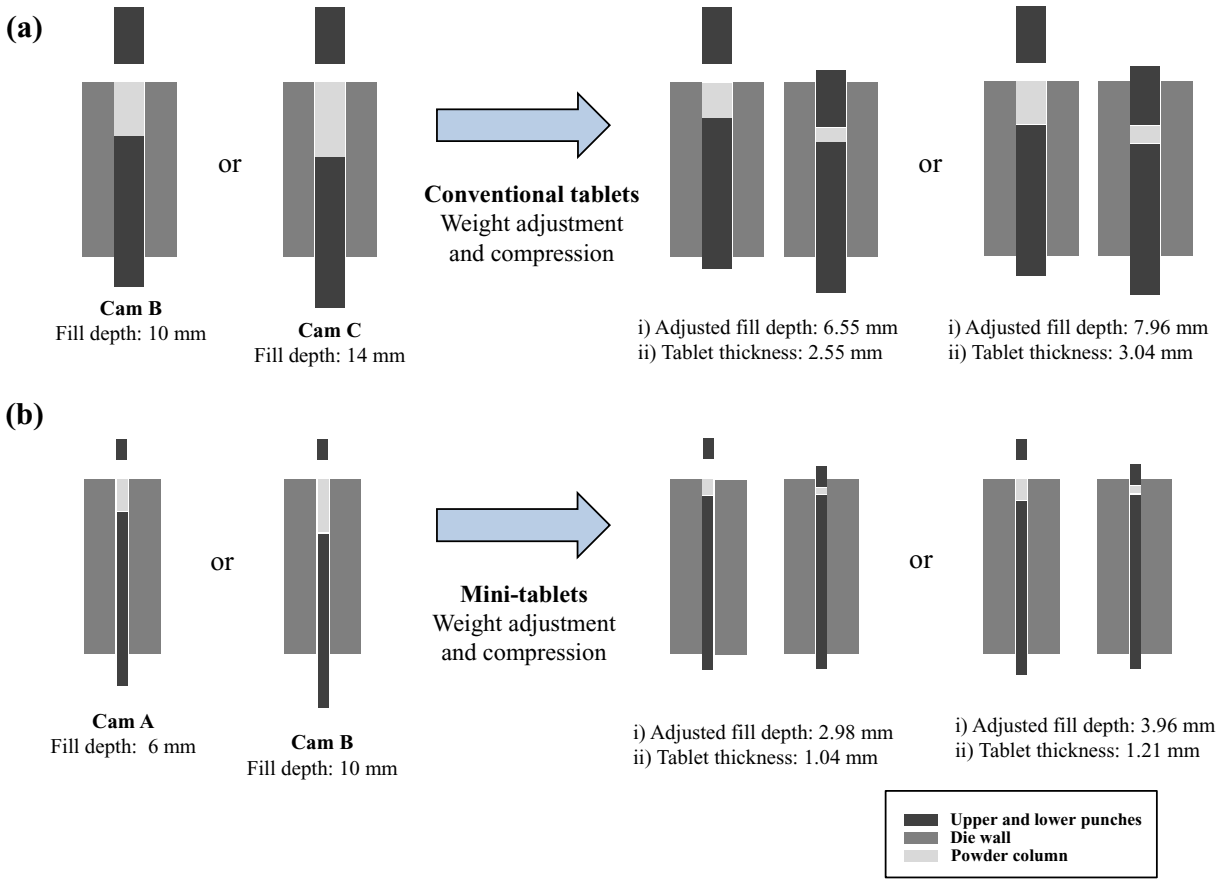


Fig. 3. Flow of the tableting process in the production of (a) conventional tablets (5.5 mm) and (b) mini-tablets (2 mm).

significantly when compared with that of initial granule, demonstrating no distinct friction occurred in feed frame (Data not shown).

3.2. Die filling behavior as a function of cam type, adjusted fill depth, and rotational speed

In this study, cam type, adjusted fill depth, and rotational speed were controlled to investigate their effects on die filling behavior during the production of conventional tablets (5.5 mm) and mini-tablets (2 mm).

3.2.1. Conventional tablets (5.5 mm)

3.2.1.1. Adjusted fill depth of 6.55 mm. As presented in Fig. 4a and c, with increasing rotational speed, the die filling efficiency for both cam types (cam B and C) was increased until reaching the critical rotational speed (45 rpm for both cam B and C). Because the ratio between the disc rotational speed and feeder paddle was kept constant throughout the experiments, the number of paddle passes over the die cavity could not be the cause of this increased powder filling. Thus, the following two possible reasons were suggested. One was the increased speed of feeder paddle, which might have forced granules to flow into the die cavity. The other was the enhanced suction effect, which was generated by the downward movement of the lower punch, creating differential pressure between the powder bed and the die cavity. However, considering the structure of the feed frame and type of the feeder paddle used in this study, it is likely that the increased feeder paddle speed had no great effect on die filling. The feed frame had a clearance of approximately 5 mm between the feeder paddle and die surface. Thus, the granules could escape from direct compression by the feeder paddle owing to their small volumetric size (shown in Table 3). Moreover, the feeder paddle used was of flat type which was reported to exert less downward force than the angled type [29]. Hence, the major effect of the feeder

Table 2
Changes in powder column thickness inside the die cavity.

Manufacturing process (tablet type/cam type/adjusted fill depth)	After overfill by fill cam	After weight adjustment	Tableting
Conventional tablets/Cam B/6.55 mm	10 mm	6.55 mm	2.55 mm
Conventional tablets/Cam C/6.55 mm	14 mm	6.55 mm	2.55 mm
Conventional tablets/Cam B/7.96 mm	10 mm	7.96 mm	3.04 mm
Conventional tablets/Cam C/7.96 mm	14 mm	7.96 mm	3.04 mm
Mini-tablets/Cam A/2.98 mm	6 mm	2.98 mm	1.04 mm
Mini-tablets/Cam B/2.98 mm	10 mm	2.98 mm	1.04 mm
Mini-tablets/Cam A/3.96 mm	6 mm	3.96 mm	1.21 mm
Mini-tablets/Cam B/3.96 mm	10 mm	3.96 mm	1.21 mm

Table 3
Particle size distribution of granules (mean ± S.D., n = 3).

Aperture size of the sieve (μm)	Retained fraction (%)
425 μm	8.879 ± 0.473
355 μm	2.935 ± 0.306
300 μm	10.427 ± 0.400
250 μm	8.668 ± 0.454
200 μm	5.873 ± 0.210
150 μm	10.846 ± 0.127
106 μm	20.671 ± 0.092
75 μm	19.984 ± 1.251
Receiver	11.716 ± 0.800
Volumetric diameters (μm)	D_{10} = 32.103 ± 2.123
	D_{50} = 123.692 ± 1.440
	D_{90} = 365.027 ± 12.403

Table 4
Flow properties of granules (mean ± S.D., n = 3).

Flow properties	Measured values	USP criteria
Bulk density (g/mL)	0.576 ± 0.004	–
Tapped density (g/mL)	0.687 ± 0.005	–
Compressibility index (%)	16.086 ± 0.074	Fair
Hausner ratio	1.192 ± 0.001	Fair
Angle of repose (°)	37.145 ± 0.383	Fair-aid not needed

paddle was to facilitate the flow of granules to the die filling zone, rather to directly force granules into the die openings. Even though increased die filling with increasing feeder paddle speed was previously reported [29], the ratio between the disc rotational speed and feeder paddle was not fixed in that case, which was different from this study (fixed). Thus, it was logical to conclude that increased die filling efficiency was mainly caused by the suction effect. Studies that proved the effect of suction on die filling [32–34] and the greater influence of suction than that of feeder velocity [26] also support this conclusion.

The increased downward velocity of the lower punch with increasing rotational speed (from 15 to 45 rpm) could result in more intensive differential pressure along the powder bed facilitating more powder to enter the die cavity, even though the retention time of the die cavity under powder bed was reduced. Because the fill depth of cam C was higher than that of cam B (10 mm for cam B; 14 mm for cam C), the greater differential pressure produced by cam C could be expected to fill more powder. However, at a rotational speed of 30 rpm, it was observed that cam B more filled the die cavity than cam C. This can be explained by the balance between the suction effect and the retention time of the die cavity under powder bed (hereinafter referred to as die retention time). As rotational speed increased, die retention time decreased while the suction effect increased. From this aspect, it appeared that the die retention time at 30 rpm was too short for granules to densely fill the die cavity, whereas the suction effect was not high enough to compensate. Thus, granules could be loosely packed along the powder column with cam C, whereas cam B could produce more compact powder column. When these two columns became the same volume after the weight adjustment process, denser powder column obtained by cam B could result in heavier column (that is, more powder filled in the same volume). However, from a rotational speed of 45 rpm, the suction effect of cam C was high enough to compensate for the compromised die filling efficiency. There was no significant difference in mean tablet weight between the two cam types (cam B and C), and the mean tablet weight almost reached a plateau (Fig. 4a and c). Considering the deeper fill depth of cam C, it can be also inferred that the die cavity with larger volume can be filled more efficiently by higher suction effect so that adjusted fill weights for both cam types become comparable.

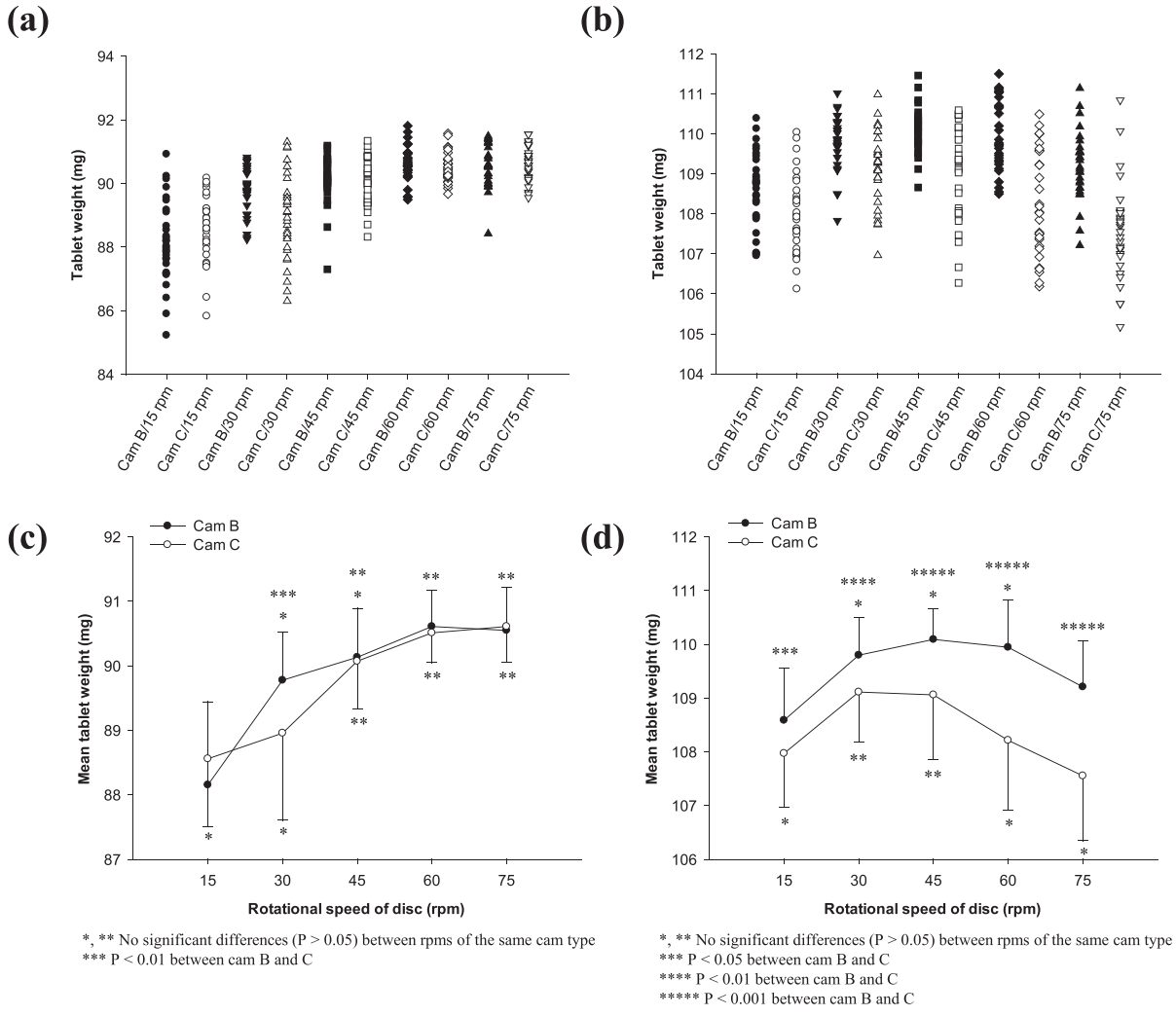


Fig. 4. Weight distribution of conventional tablets (5.5 mm) with adjusted fill depths of (a) 6.55 mm and (b) 7.96 mm as a function of rotational speed ($n = 30$). Mean weight of conventional tablets with fill depths of (c) 6.55 mm and (d) 7.96 mm as a function of rotational speed (mean ± S.D., n = 30).

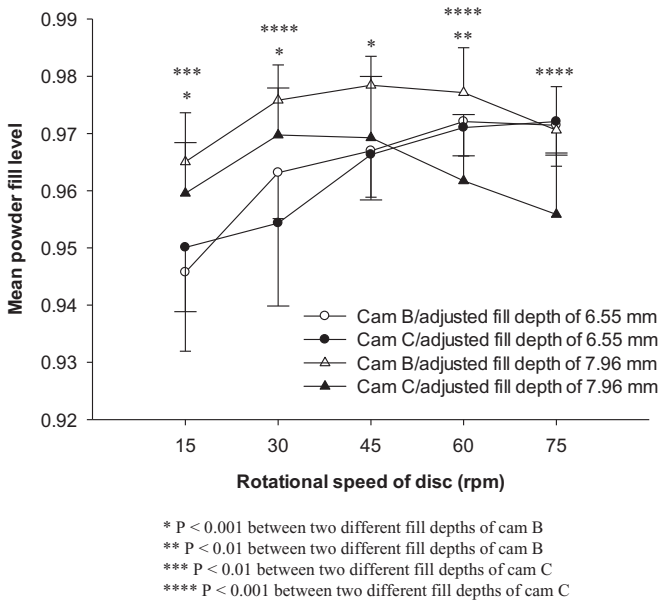


Fig. 5. Mean powder fill level of conventional tablets (5.5 mm) as a function of rotational speed (mean ± S.D., n = 30). Error bars with longer horizontal tips denote the larger adjusted fill depth of the same cam type.

In the meantime, as presented in Fig. 5, the mass of the powder filled inside the die cavity reached more than 96% of the theoretical fill mass at a rotational speed of 45 rpm for both cam types, and it did not significantly increase further. Hence, it can be postulated that the suction effect enhanced by the high rotational speed could increase die filling efficiency to a limited extent in a given condition. For more powder fill, adjusting the other parameters, such as adjusted fill depth, geometry of die and lower punch, friction between granules and die wall, and shape or size distribution of granules, can be considered as an option.

3.2.1.2. Adjusted fill depth of 7.96 mm. As shown in Fig. 4b and d, when the adjusted fill depth was increased to 7.96 mm, the mean tablet weights with cam B were significantly higher than those with cam C at all rotational speeds (15, 30, 45, 60, and 75 rpm). Because the die retention time was the same for both cam B and C at the same rotational speed, it could be concluded that these results were caused by the insufficient suction effects. That is, if the suction effect had compensated for the increased filling requirement (change in cam type from B to C), the density of powder columns after passing the fill cam could have been similar resulting the mean tablet weights for both cam types comparable. Therefore, it was logical to say that the increased suction effect by deeper fill depth (cam C) was not high enough to fill the die cavity of increased volume with similar powder density.

In contrast to the case of adjusted fill depth of 6.55 mm, there was a critical rotational speed for both cam types (cam B and C) at which the die filling efficiency significantly decreased (75 rpm for cam B; 60 rpm for cam C). Again, this can be explained by the balance between the decreased die retention time and the increased suction effect as the function of increased rotational speed. The die retention time at rotational speeds of 60 and 75 rpm was relatively short to be compensated by the enhanced suction effect.

Meanwhile, the mean powder fill level at high rotational speed also decreased with decreasing mean tablet weight as shown in Fig. 5.

3.2.1.3. Comparison between adjusted fill depths of 6.55 and 7.96 mm. From the results of mean powder fill level (Fig. 5), it was found that the initial powder column produced after passing the die fill region of rotary tablet press had non-homogeneous density distribution. This is plausible since the tablet weights were different according to the adjusted fill depth (6.55 or 7.96 mm) even though they were generated from the identical initial powder columns which were produced by the same cam type. Thus, the density distribution for initial powder column according to the cam type and rotational speed could be suggested as Fig. 6 where the gradient of powder density was observed. In both cam types (Fig. 6a and b), the upper section of the initial powder column (colored as black) was found to be denser than the lower section (colored as light gray) at the rotational speeds of 15 and 30 rpm. However, with cam C (Fig. 6b), the two densities of the upper and lower sections of the initial powder column became similar at 45 rpm and then inversed from 60 rpm, while they became similar at 75 rpm with cam B (Fig. 6a). These trends according to the rotational speed could be explained by the suction effect which has the potential to destabilize arches and facilitate flow, as reported by a previous research [35]. As the differential pressure along the powder bed

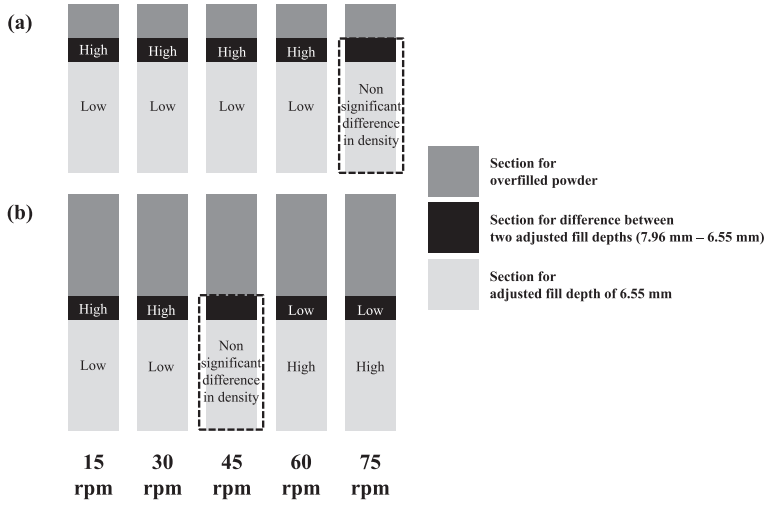


Fig. 6. Representative scheme showing the density distribution of powder column after passing (a) cam B and (b) cam C at various rotational speeds for producing conventional tablets (5.5 mm).

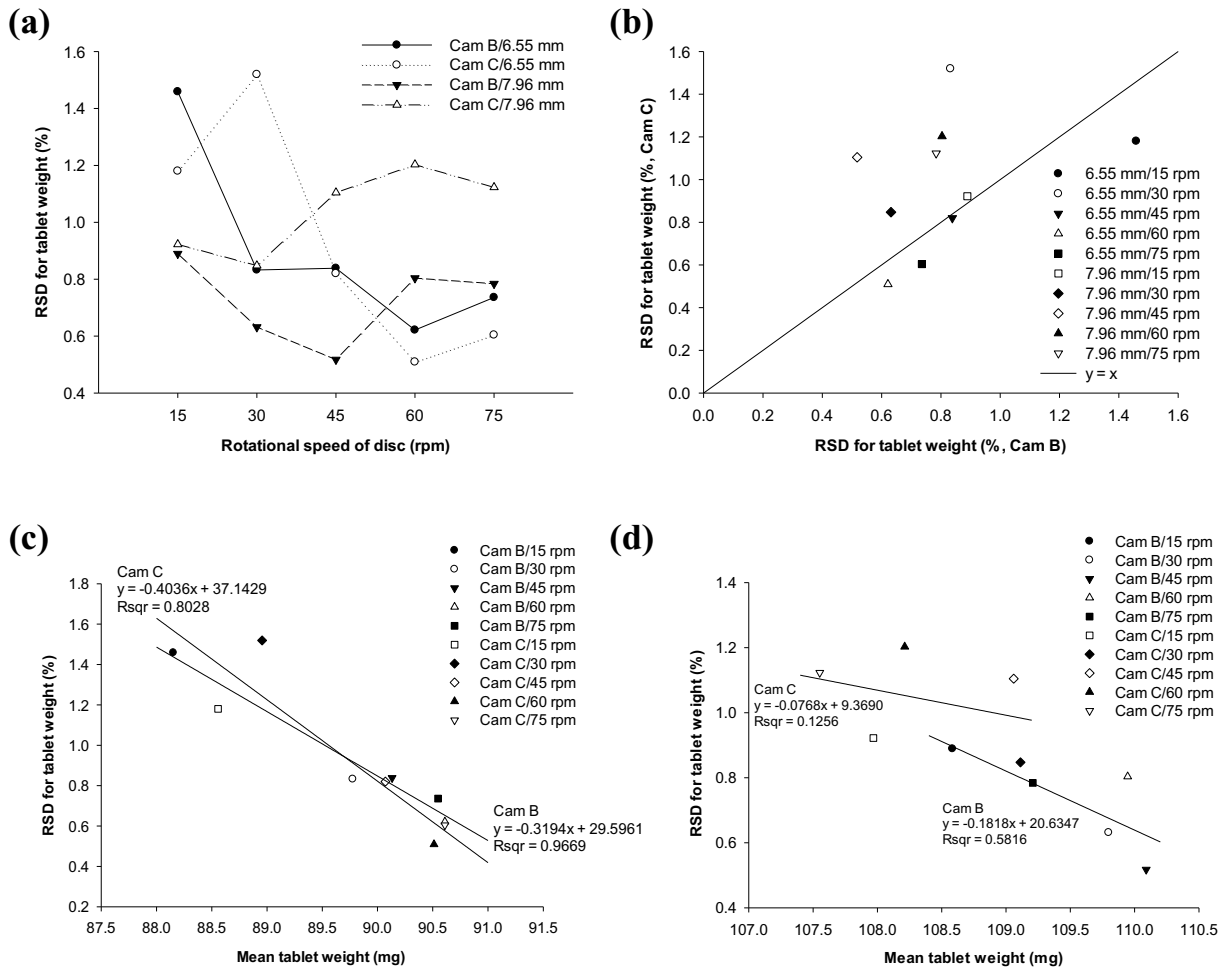


Fig. 7. (a) RSD for conventional tablet weights (5.5 mm) as a function of cam type, adjusted fill depth, and rotational speed. (b) RSD distribution of conventional tablet weights according to the production process. A plot of the RSD of conventional tablet weights against mean weight of conventional tablets with adjusted fill depth of (c) 6.55 mm and (d) 7.96 mm.

above die cavity became higher with increasing rotational speed, powder could be deposited deeper. On the contrary, at the lower rotational speed, the relative influence of gravity on die filling behavior could be higher compared with that of the higher rotational speed, diminishing the relative influence of powder suction. That is, the suction force was not high enough to deliver the powder to the bottom of the powder column.

On the other hand, early inversion of powder density in initial powder column was observed in cam C. Again, this could be attributed to the higher suction force which has potential to break the localized arches inside the die cavity, transporting more powder to the bottom. Inaddition, the increased centrifugal force at high rotational speed might have contributed to the density inversion.

3.2.1.4. Relative standard deviations for conventional tablets. As shown in Fig. 7a and b, relative standard deviation (RSD) of tablet weight had a tendency to decrease by cam B than cam C for most cases of conventional tablet production. To elucidate the results, a regression analysis was conducted for the RSD and mean tablet weight according to the cam type and adjusted fill depth (Fig. 7c and d). In general, there is inevitable deviation for the die filling process because reduction in granule size is limited to an extent. Therefore, it is expected that the increasing tablet weight usually decreases the RSD in tablet weight. From this perspective, it was concluded that the high R-squared values of regression models for both cam types with the adjusted fill depth of 6.55 mm (Fig. 7c, 0.9669 for cam B and 0.8028 for cam C) indicates that the main cause of the decrease in RSD was attributed to the increased tablet weight. Moreover, the tendency of decrease in RSD with increasing rotation speed (Fig. 7a) supports this conclusion since the tablet weights were increased by the enhanced suction effect as demonstrated in Fig. 4a and c. Meanwhile, the relatively lower R-squared value of regression model for cam C with the adjusted fill depth of 6.55 mm could be resulted from the more dynamic die filling behavior occurring inside the cam C due to the higher suction effect (Fig. 6b). The increased air entrapment could contribute to these higher variations in die filling.

On the contrary, with the processes using the adjusted fill depth of 7.96 mm, the goodness of fit significantly decreased for both cam types (Fig. 7d R-squared value of 0.5816 for cam B and 0.1256 for cam C), while the process of cam B showed more similar trend with the results of the adjusted fill depth of 6.55 mm (Fig. 7c) compared with that of cam C. Meanwhile, it was found that the overall RSD for the processes with cam B and adjusted fill depth of 7.96 mm was lower than that of 6.55 mm (Fig. 7a and b). In addition, the RSD for adjusted fill depth of 7.96 mm was lower with cam B than with cam C at all rotational speed (Fig. 7a and b). These suggest that the process applying cam B and adjusted fill depth of 7.96 mm enabled more reproducible die filling. Therefore, the lowered correlation between RSD and mean tablet weight with cam B applying the

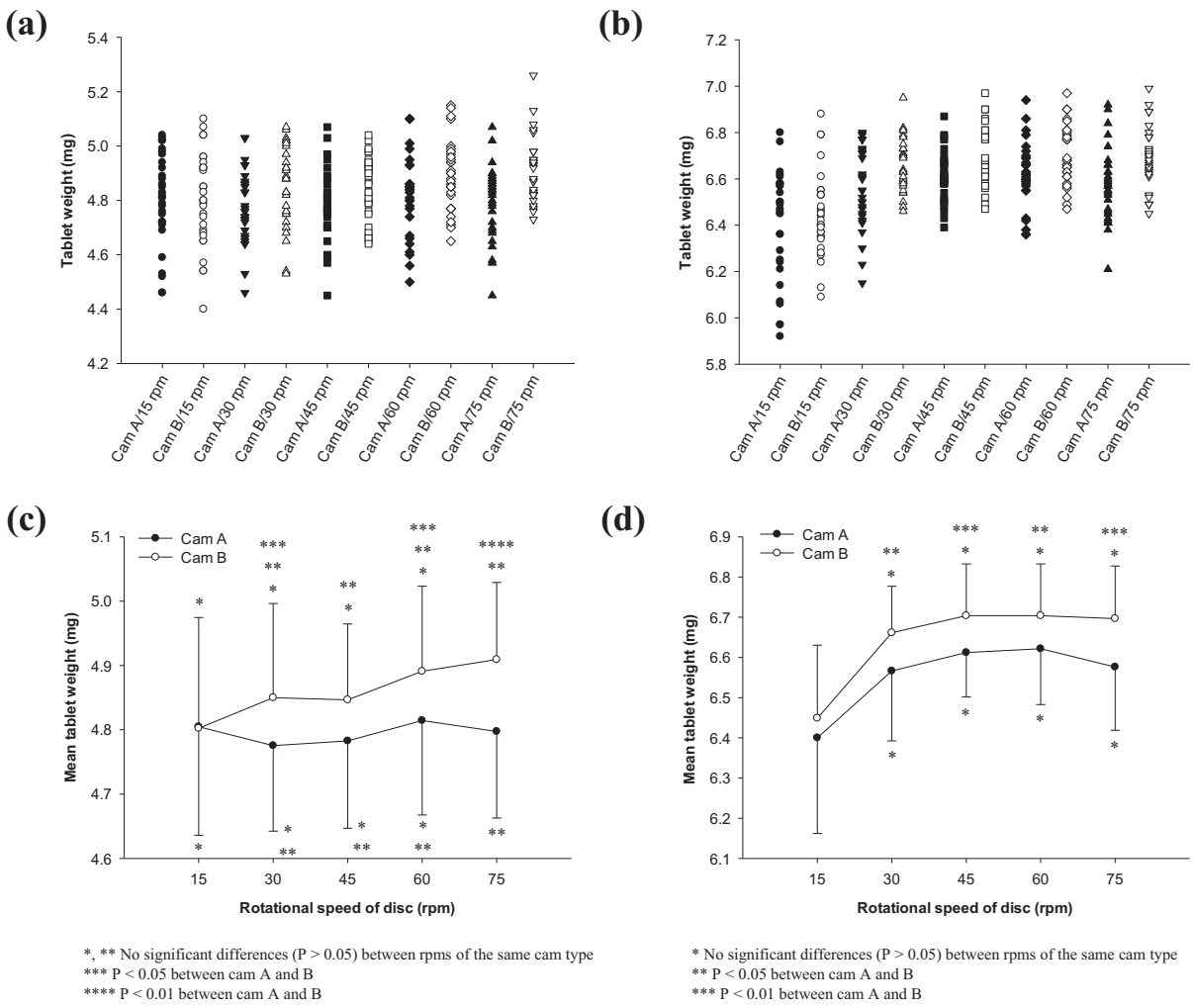


Fig. 8. Weight distribution of mini-tablets (2 mm) with adjusted fill depths of (a) 2.98 mm and (b) 3.96 mm as a function of rotational speed (n = 30). Mean weight of conventional tablets with fill depths of (c) 2.98 mm and (d) 3.96 mm as a function of rotational speed (mean ± S.D., n = 30).

adjusted fill depth of 7.96 mm was attributed to the inevitable deviation in the given process, which was difficult to be improved simply by increasing the fill weight. During the die filling with the cam C and adjusted fill depth of 7.96 mm, it was more difficult to overcome the inevitable deviation of the given process condition since more dynamic die filling behavior was achieved by higher suction effect compared with cam B (Fig. 6b).

3.2.2. Mini-tablets (2 mm)

3.2.2.1. Adjusted fill depth of 2.98 mm. As presented in Fig. 8a and c, when cam A was applied, it was difficult to increase die filling efficiency by adjusting the rotational speed. On the contrary, cam B slightly increased die filling efficiency at a rotational speed of 75 rpm, which was attributed to enhanced powder suction. At rotational speeds ranging from 30 to 75 rpm, cam B produced heavier tablets compared with cam A, which was different from the case of conventional tablet production where the cam with lower fill depth had tendency to fill the die cavity more efficiently than that with deeper fill depth. Therefore, it was logical to rule out the effects of die retention time from possible reason for these results. Accordingly, it was concluded that the suction effect played a crucial role during the manufacturing of mini-tablets. In other words, the enhanced suction effect by the faster downward movement of the lower punch in cam B resulted in the improved die filling efficiency at 75 rpm, while the shallow fill depth of cam A was not suitable for enhancing die filling efficiency owing to its insufficient suction effect.

Meanwhile, the overall mean powder fill levels of mini-tablets with the adjusted fill depth of 2.98 mm were far lower than those of conventional tablets (Figs. 5 and 9). With larger die diameter (5.5 mm), the ratio of "median volumetric diameter of granules to diameter of die opening" was 0.0277, whereas that with smaller diameter (2 mm) was 0.0762. Thus, there was no doubt that relatively free and smooth bulk flow of granules into the die cavity would occur during the conventional tablet production. On the contrary, flow problems, such as arch formation by the interlocking between granules, might have occurred more frequently inside the die cavity during the mini-tablet production. Accordingly, external forces such as suction were necessary for enhancing die filling efficiency for mini-tablets. The previous study [35] also suggested a need for differential pressure to break powder arches and maintain flow during the process using small orifices.

3.2.2.2. Adjusted fill depth of 3.96 mm. Fig. 8b and d show that die filling efficiency almost became constant at a rotational speed of 30 rpm for both cam A and B. However, there was no statistically significant

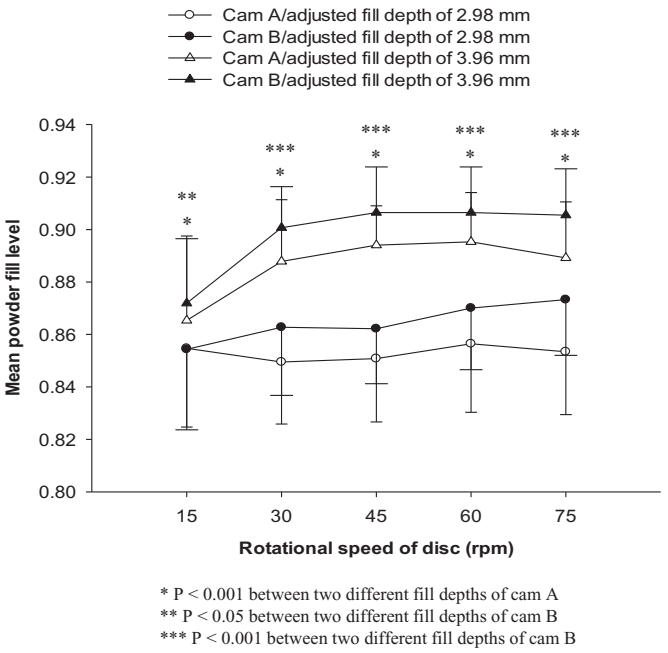


Fig. 9. Mean powder fill level of mini-tablets (2 mm) as a function of rotational speed (mean ± S.D., n = 30). Error bars with longer horizontal tips denote the larger adjusted fill depth of the same cam type.

decrease in die filling efficiency after reaching plateau, which was observed during the production of conventional tablets with the adjusted fill depth of 7.96 mm (Fig. 4b and d). This was because the die filling efficiency by gravity fill was relatively lower when producing mini-tablets due to flow problems like arch formation. That is, with a die with larger diameter (5.5 mm), the die cavity could be efficiently filled by the spontaneous flow of granules even at lower rotational speed. Accordingly, the decreased die retention time by the increased rotational speed could detrimentally affect the mean powder fill level. However, with a die with smaller diameter (2 mm), the suction force was required to maximize powder fill into the thin and long-shaped die cavity. Thus, it was evident from this study that suction effect was the main parameter affecting die filling behavior during the mini-tablet production.

In the meantime, mean powder fill level also increased as mean tablet weight increased (Fig. 9).

3.2.2.3. Comparison between adjusted fill depths of 2.98 and 3.96 mm. Fig. 9 shows that the mean powder fill levels were always higher with the fill depth of 3.96 mm than with that of 2.98 mm. The deduced results of powder density distribution shown in Fig. 10 suggested that the powder density of upper section of the initial powder column (colored as black) was always denser than that of lower section (colored as light gray). This was different from the case of conventional tablet production where even the inversion of powder density was observed with increasing rotational speed. This could be attributed to the less suitable condition for suction behavior to fully exhibit its effect on die filling (Figs. 8 and 9) due to the thin and long shape of the die cavity. This is in accordance with a previous report [26] where the suction effect was more effective with a larger die orifice than with a smaller one.

3.2.2.4. Relative standard deviations for mini-tablets. As shown in Fig. 11a and b, in most cases of mini-tablet production, lower RSD for tablet weight was obtained when cam B was applied regardless of the adjusted fill depth, indicating that the cam B enabled a more reproducible die filling. With cam A and adjusted fill depth of 2.98 mm, decreased RSD in tablet weight (black solid line in Fig. 11a and dashed square in Fig. 11c) without significant changes in tablet weight was observed when the rotational speed was increased from 15 to the higher speeds (30, 45, 60, and 75 rpm). This was attributed to the balance between the decreased die retention time and the increased suction effect. That is, the possibility of intermittent powder flow which is mainly occurred by gravity was reduced, whereas more-reproducible die filling could be accomplished by the enhanced suction force which can facilitate the arrangement of powders to specific direction. However, since there was no evident trend between the RSD and mean tablet weight, the R-squared value indicated the lack of fit (0.3641). In contrast, with cam B and adjusted fill depth of 2.98 mm, the goodness of fit significantly increased (R-squared value of 0.6142), suggesting a better linear relationship between the RSD and mean tablet weight due to the enhanced powder suction by deeper fill depth. The lower correlation between RSD and mean tablet weight for the processes of mini-tablet production (Fig. 11c) compared with that of conventional tablet (Fig. 7c) was mainly resulted from the inevitable die fill variation during the process. In other words, the process utilizing relatively small amount of powder for die filling (less than 5 mg for mini-tablet production with the adjusted fill depth of 2.98 mm, approximately 90 mg for conventional tablet production with the adjusted fill depth of 6.55 mm) was more vulnerable to weight variations.

With the adjusted fill depth of 3.96 mm, the correlation between the RSD and mean tablet weight became higher in both cam types (Fig. 11d R-squared value of 0.9091 for cam A and 0.8786 for cam B), indicating that weight increase was the main reason for the decreased RSD. As the amount of powder to be filled increased, the inevitable die fill variation during the process decreased (Fig. 11a and b).

In summary, the results above indicated that the increased tablet weight by the enhanced suction and the diminished influence of gravity

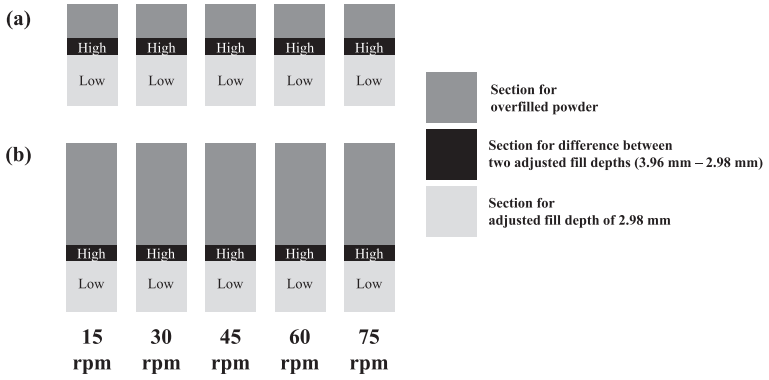


Fig. 10. Representative scheme showing the density distribution of powder column after passing (a) cam A and (b) cam B at various rotational speeds for producing mini-tablets (2 mm).

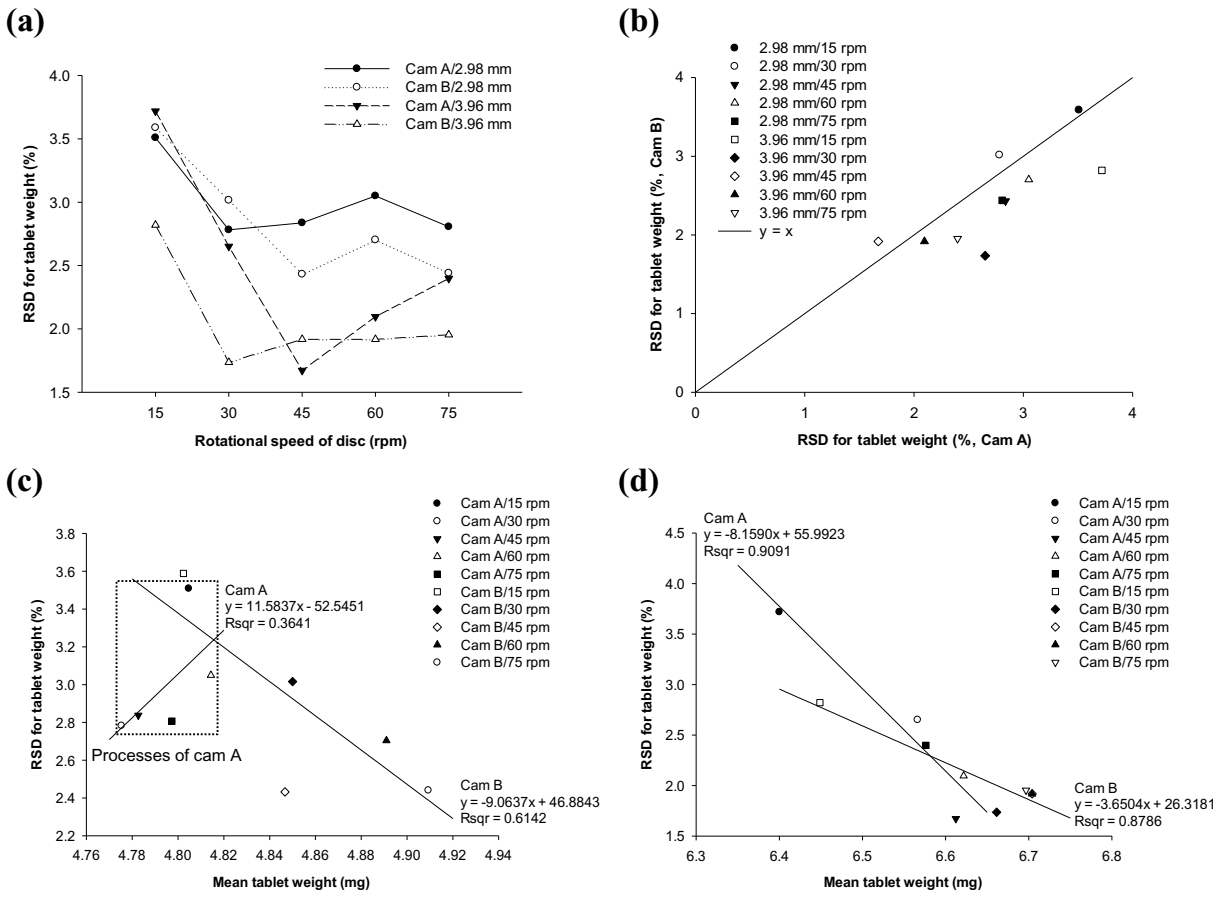


Fig. 11. (a) RSD for mini-tablet (2 mm) weights as a function of cam type, adjusted fill depth, and rotational speed. (b) RSD distribution of mini-tablet weights according to the production process. A plot of the RSD for mini-tablet weights to mean weight of mini-tablets with adjusted fill depth of (c) 2.98 mm and (d) 3.96 mm.

fill by the reduced die retention time concurrently contributed to the decrease in RSD.

4. Conclusions

To the best of our knowledge, this is the first practical study to compare the die filling behavior of conventional tablets and mini-tablets by modifying cam type, adjusted fill depth, and rotational speed on a rotary tablet press. During the production of conventional tablets, the shallow fill cam and adequate rotational speed were suitable for an efficient and reproducible process because the granules spontaneously flowed into the die cavity without a critical problem. Thus, it was demonstrated that balancing the die retention time and suction effect was key in improving die filling behavior in the production of conventional tablets. In contrast, for the mini-tablets, maximizing suction effect using deeper fill cam and high rotational speed was required to put more powder into the die cavity and reduce deviation in tablet weights. However, the quality attributes of tablets, such as dissolution, and mechanical properties, should also be considered since they are closely related to the process parameters.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.powtec.2019.11.109.

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